

EPS261 Final Assignment Report

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A pseudo-spectral method is used to solve the sea level equation (1) without considering shoreline migration, when Greenland or West Antarctica ice is melted uniformly by unit 1 (ice change shown in Fig. 1,2). The maximum degree is 256 in this calculation and three iterations are done to both fingerprints.

$$\left\{ \begin{array}{l} \Delta S(\theta, \psi) = \left[\Delta \mathcal{SL}(\theta, \psi) + \frac{\Delta \Phi}{g} \right] C(\theta, \psi) \\ \frac{\Delta \Phi}{g} = \frac{1}{A} \frac{\rho_i}{\rho_w} \iint_{\Omega} \Delta I(\theta, \psi) d\Omega - \frac{1}{A} \iint_{\Omega} \Delta \mathcal{SL}(\theta, \psi) C(\theta, \psi) d\Omega \\ \Delta \mathcal{SL}(\theta, \psi) = \sum_{l,m} T_l E_l (\rho_i \Delta I_{lm} + \rho_w \Delta S_{lm}) Y_{lm}(\theta, \psi) \end{array} \right. \quad (1)$$

The resultant fingerprint for each glacier melting is shown by Figure 3 and 4 (re-scaled so the global mean is one for each fingerprint). The feature that sea level near the melting ice is decreasing while far-field sea level is rising has been recovered.

If the calculated fingerprint is used to reconstruct the sea level change measured by tide gauges whose positions are shown in Figure 5. The least square fit of the Greenland fingerprint and Antarctica fingerprint to match the measurements gives that the observations contain 0.45 from the scaled Greenland fingerprint and 1.22 from the scaled Antarctica fingerprint (Fig. 6).

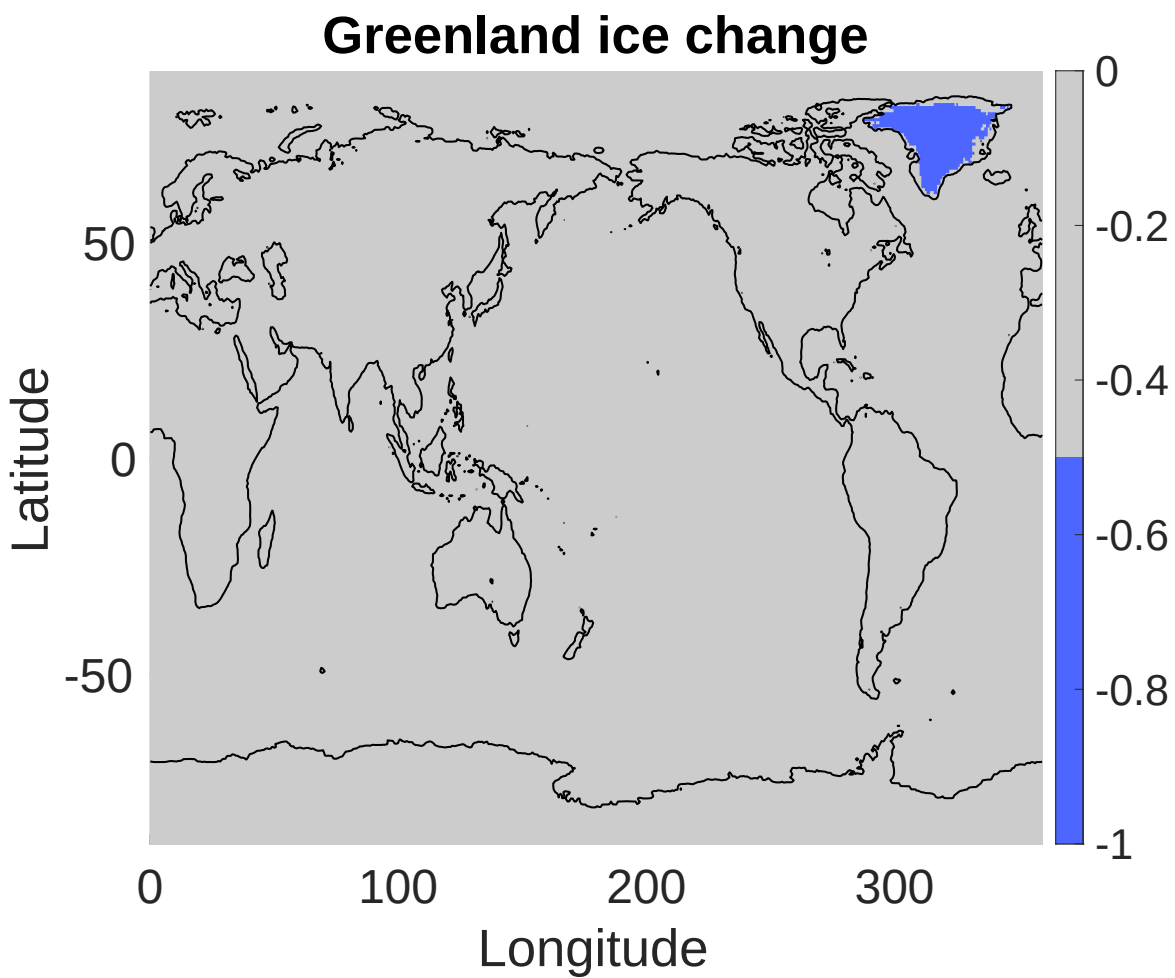


Figure 1: Unit ice melting on Greenland

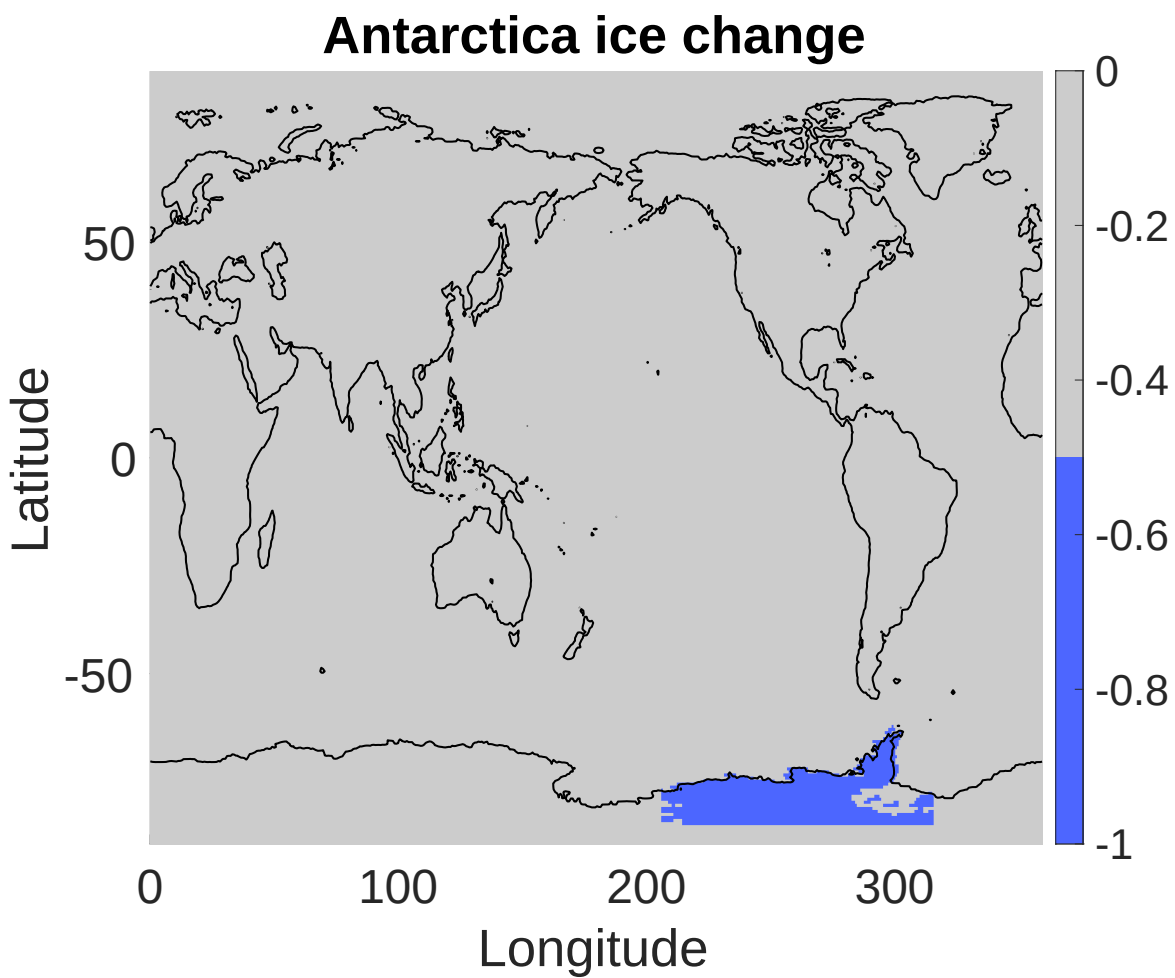


Figure 2: Unit ice melting on Antartctica

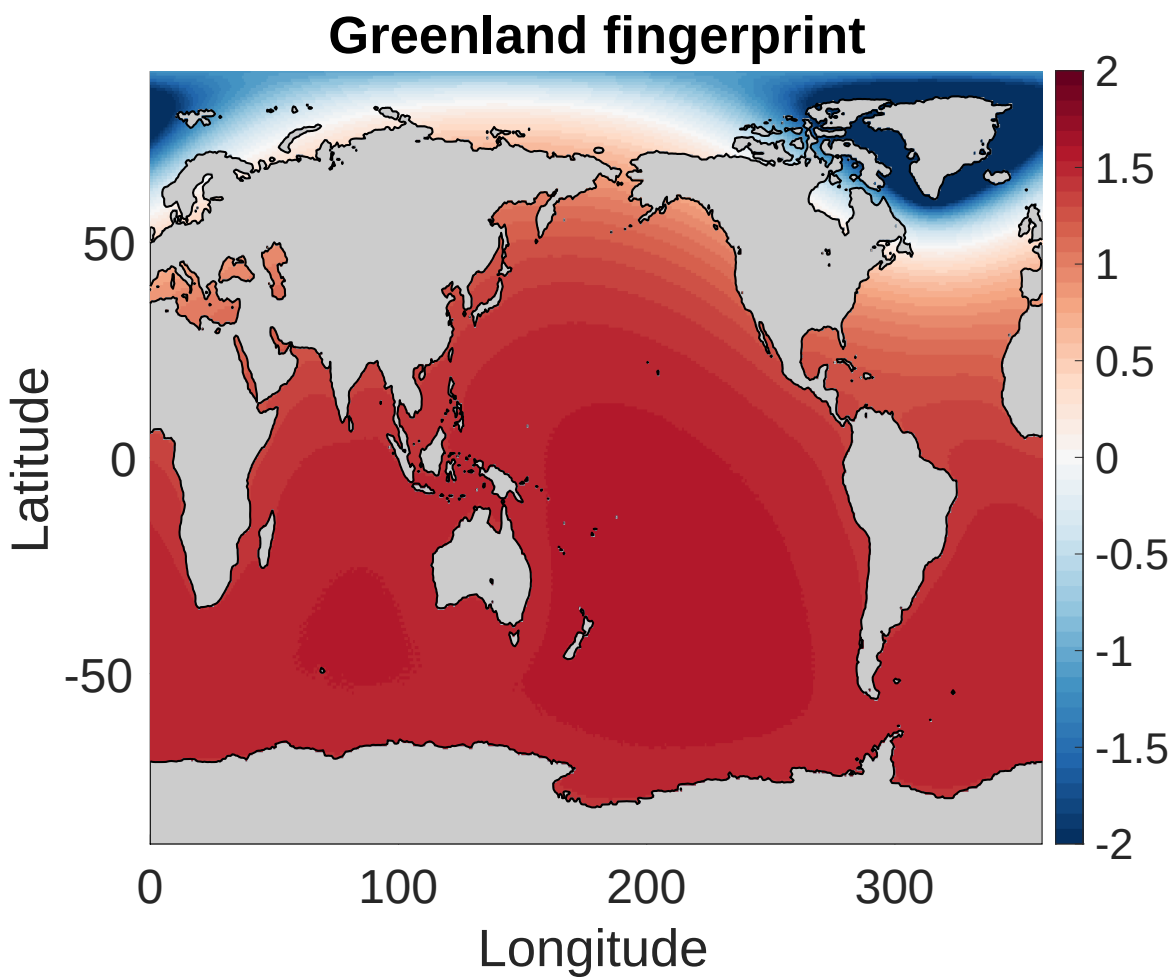


Figure 3: Greenland fingerprint (re-scaled so global average is one)

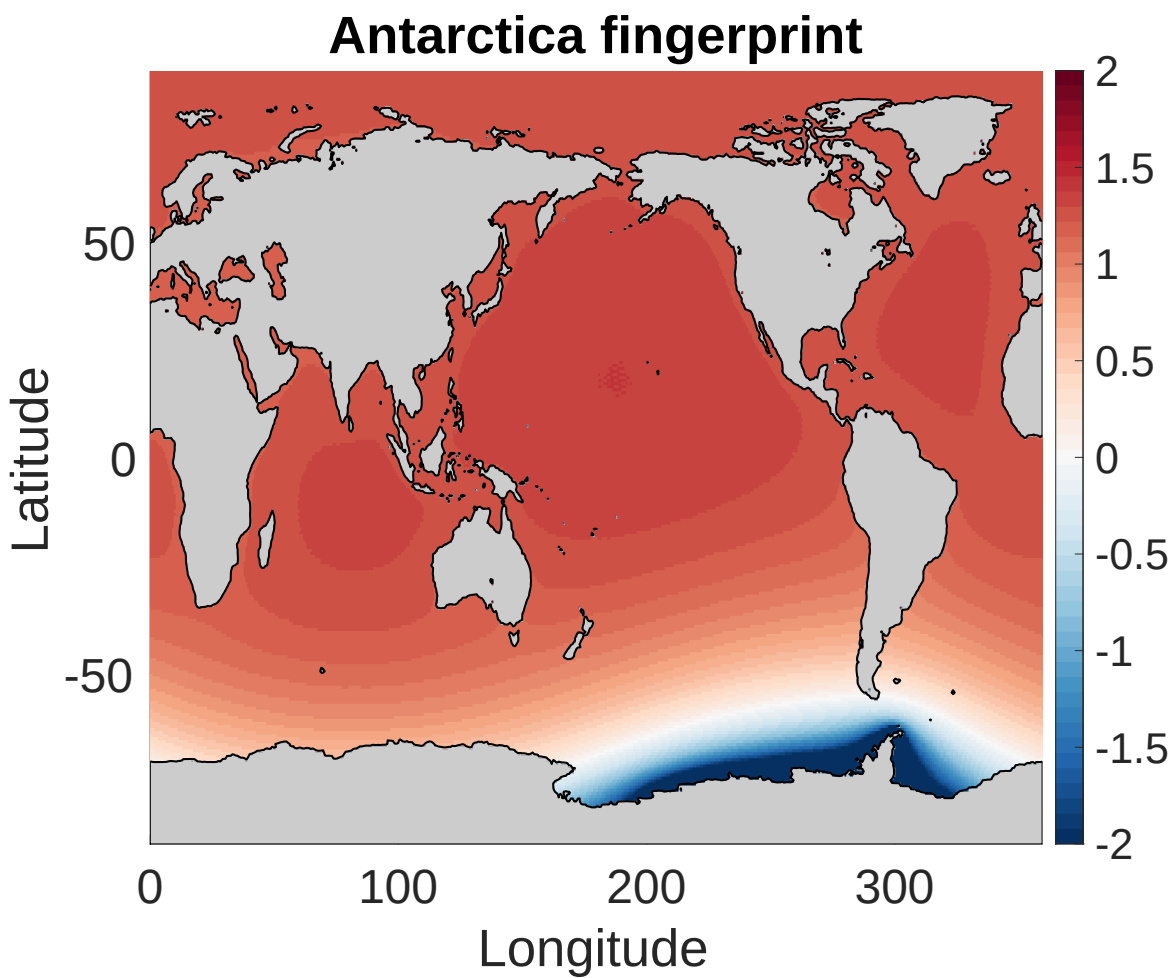


Figure 4: Antarctica fingerprint (re-scaled so global averaged is one)

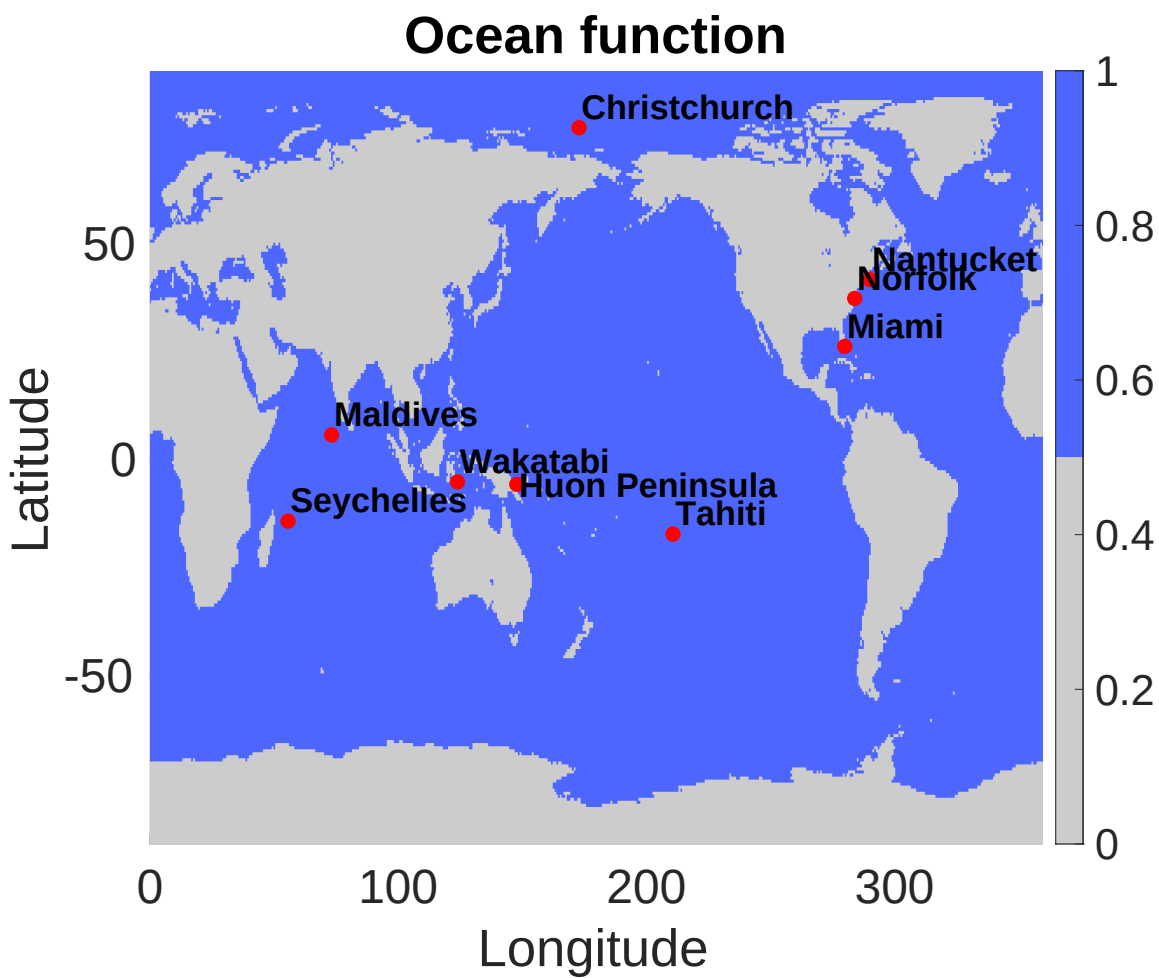


Figure 5: Ocean mask (blue), continents (gray), and locations of the tide gauges (red dots)

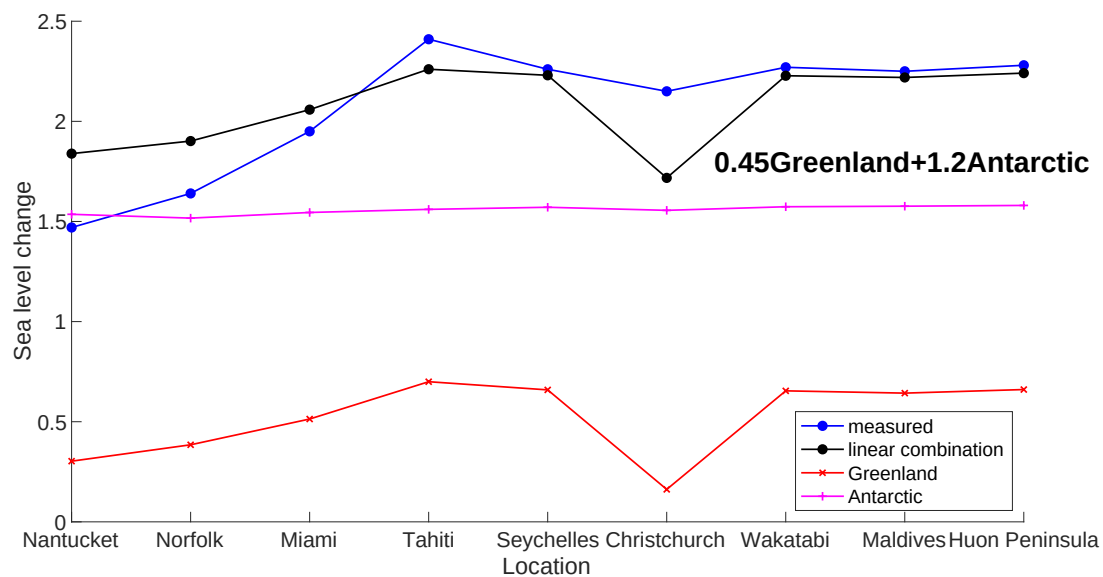


Figure 6: Least square fit of fingerprints to tide gauge observations. Blue line: observations; black line: least square fit; red line: Greenland contribution; magenta line: Antarctica contribution.